TO 33kV SWITCHBOARD F00-SW-010 PANEL B20
REFER 2000-F00-SLD-EL-10005

F16-NR-011
FLOTATION TRANSFORMER F16-TR-011
NEUTRAL EARTHING CUBICLE

INTERTRIPS TO SIGNALS

F16-TR-011
33 /11 kV
10000 kVA
Dyn11
z=8%
KNAN

254Ω +/-5%
30/0.05A 25A FOR 10s
SENSING
ER-15kV

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

F16-SR-001
11kV FLOTATION SWITCHROOM

F16-SW-011
FLOTATION 11kV SWITCHBOARD

NOTE 4

E-SHUT
F16SR001-LCS
NOTE 5
TO GENERATOR INCOMER

TEST
600/1A 2.5VA Cl 0.2
PQM SEL735
PR SE-330AU
EARTH FAULT INDICATION
TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)
UPSTREAM TRIPS
600/1A 2.5VA 5P20
CLOSE
TRIP
1250A
NOTE 2
50 50G 51G 50SG AFD 50N 51N PR SEL 751 26Q 49W 63 71
NOTE 3

TO BUS EARTH SWITCH/VT
11kV 1250A 3-E 25kA FOR 3s
110V AC SUPPLY

A07 A08 A09 A10 A11 A12

2000-F16-SCM-EL-10117
2000-F16-SCM-EL-10123

11kV 1250A 3-E 25kA FOR 3s

100E CLOSE 50 50G 51 51G 81 AFD PR SEL 751 NOTE 3
160E CLOSE 50 50G 51 51G 81 AFD PR SEL 751 NOTE 3
630A CLOSE TRIP 50 50G 51 51G 81 AFD PR SEL 751 NOTE 3

50/1A 2.5VA 5P20 TEST
50/1A 1.5VA 10P10 TEST
VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT

F16AG426-VS
SCAVENGER FLOTATION CELL 3 VSD
11/3.3kV 500kW

F16AG427-VS
SCAVENGER FLOTATION CELL 4 VSD
11/3.3kV 500kW

F24PW401-VS
MILL WATER PUMP 1 VSD
11/3.3kV 800kW

F24PW402-VS
MILL WATER PUMP 2 VSD
11/3.3kV 800kW

M 3~ RTD
500 kW
F16-AG-426
SCAVENGER FLOTATION CELL 3 AGITATOR
2000-F16-SCM-EL-10113

M 3~ RTD
500 kW
F16-AG-427
SCAVENGER FLOTATION CELL 4 AGITATOR
2000-F16-SCM-EL-10115

M 3~ RTD
800 kW
F24-PW-401
MILL WATER PUMP 1
2000-F16-SCM-EL-10119

M 3~ RTD
800 kW
F24-PW-402
MILL WATER PUMP 2
2000-F16-SCM-EL-10121

11kV SPARE
FULLY EQUIPPED
-

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 15/10/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND UPSTREAM 33kV FEEDER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.

LEGEND

- PR — INCOMER / FEEDER PROTECTION RELAY
- PQM — POWER QUALITY METER
- ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
- CAPACITIVE VOLTAGE INDICATION
- MECHANICAL INTERLOCK DEVICE
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- M — MOTOR OPERATOR
- 26Q — TRANSFORMER PROTECTION DEVICES
- 27 — UNDER VOLTAGE
- 38 — BEARING RTD TEMPERATURE
- 49W — TRANSFORMER WINDING TEMPERATURE
- 50 — INSTANTANEOUS OVERCURRENT
- 50SG — INSTANTANEOUS OVERCURRENT SENSITIVE GROUND
- 50G 50N — INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
- 51 — TIMED OVER-CURRENT
- 51G 51N — TIMED EARTH/NEUTRAL OVERCURRENT
- 59 — OVER VOLTAGE
- 63 — TRANSFORMER PRESSURE / BUCCHOLZ
- 64 — EARTH FAULT
- 71 — TRANSFORMER LIQUID / GAS LEVEL
- 81 — FREQUENCY
- AFD — ARC FLASH DETECTION

1CONT'D FROM SHEET 2 DRG 2000-F16-SLD-EL-10001
CONT'D ON SHEET 3 DRG 2000-F16-SLD-EL-10003

2000-F16-SLD-EL-10002_Rev2
Single line diagram sheet 2

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